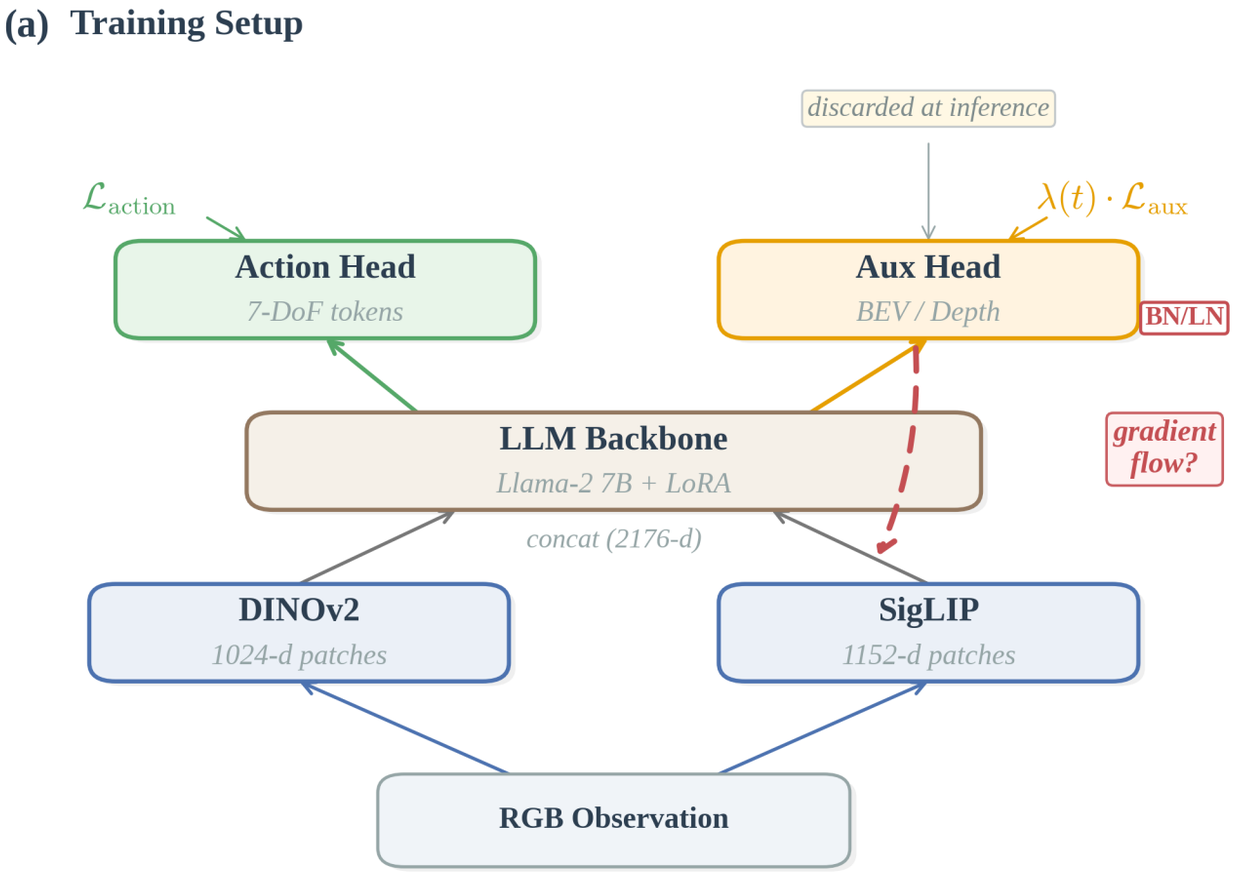
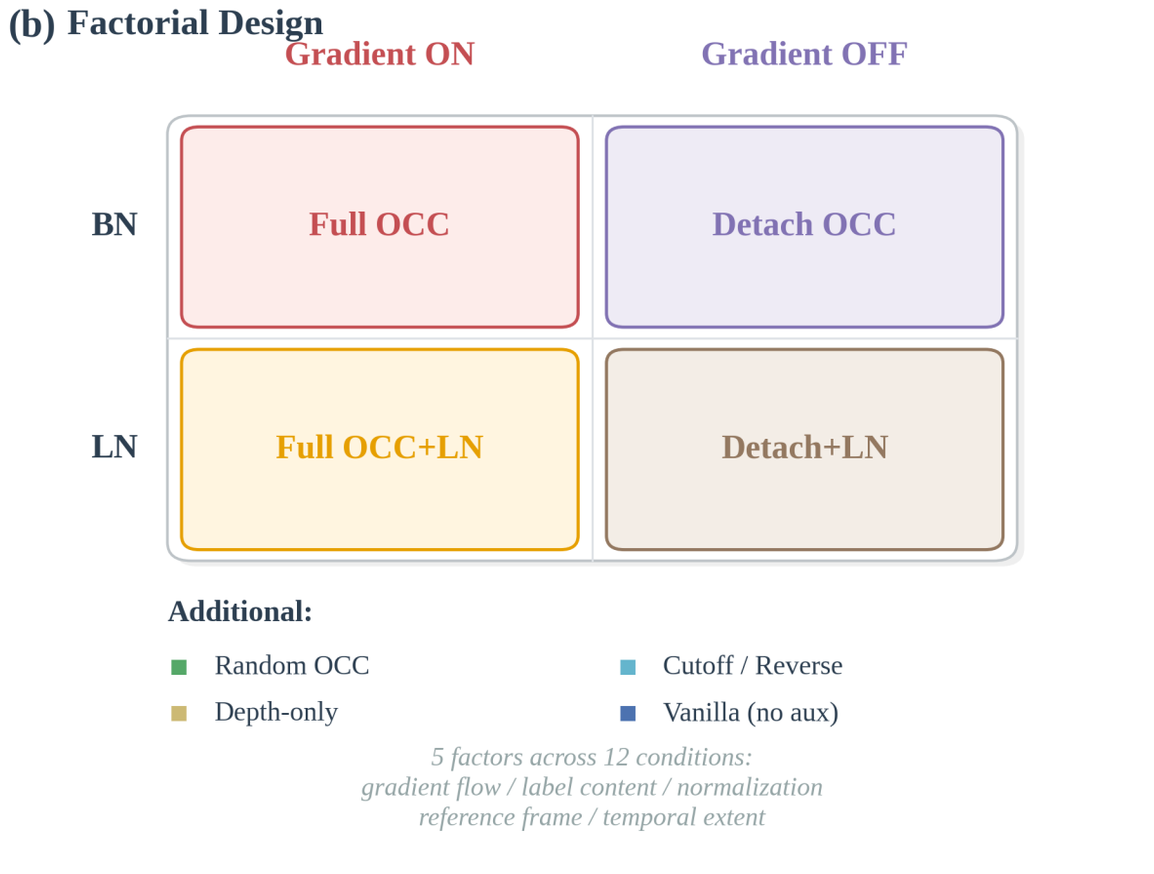

(a) Training Setup
discarded at inference
$\mathcal{L}_{\text{action}}$
$\lambda(t) \cdot \mathcal{L}_{\text{aux}}$
Action Head
7-DoF tokens
Aux Head
BEV / Depth
BN/LN
LLM Backbone
Llama-2 7B + LoRA
gradient flow?
concat (2176-d)
DINOv2
1024-d patches
SigLIP
1152-d patches
RGB Observation


(b) Factorial Design
Gradient ON
Gradient OFF
BN
LN
Full OCC
Detach OCC
Full OCC+LN
Detach+LN
Additional:
Random OCC
Cutoff / Reverse
Depth-only
Vanilla (no aux)
5 factors across 12 conditions:
gradient flow / label content / normalization
reference frame / temporal extent